



Differential relationships between sub-traits of BIS-11 impulsivity and executive processes: An ERP study

Julia W.Y. Kam, Rachelle Dominelli, Scott R. Carlson *

Department of Psychology, University of British Columbia, 2136 West Mall, Vancouver, British Columbia, Canada V6T 1Z4

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ABSTRACT

There is mixed evidence for a relationship between impulsivity and executive functions. Although impulsivity is heterogeneous, previous research did not examine partial relationships controlling for shared variance across sub-traits to evaluate the specificity of these associations. Eighty-five undergraduates completed the Barratt Impulsiveness Scale-11 (BIS-11) and the AX-expectancy version of the Continuous Performance Task (AX-CPT). This task engenders a conflict between two response tendencies by manipulating the frequency of specific trial types. We conducted mixed model analyses to determine the unique variance in behavioral and electrophysiological indices of relevant cognitive functions accounted for by the facets of BIS-11. Motor Impulsiveness was associated with smaller P3 across sites and conditions suggesting a general cognitive limitation not specific to the condition requiring the most inhibition, and larger N2 in some conditions indicating heightened conflict detection. Non-Planning Impulsiveness was related to smaller N2 when inhibiting a primed response and with greater P3 in some contexts. Attentional Impulsiveness appeared to be associated with an inefficient conflict detection system indicated by relatively normal engagement in trials involving the non-potent response, but relatively over engagement in the prepotent condition. Our findings suggest that sub-traits of impulsivity are differentially related to executive processes.

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1. Introduction

Self-regulation is fundamental to trait conceptualizations of impulsivity. Relative problems with higher order functions influencing capacity for self-control, such as “top-down” attention (e.g., [Posner and Rothbart, 2007](#)) and prefrontally-mediated inhibitory processes (e.g., [Miller and Cohen, 2001](#); [Shimamura, 2000](#)), may serve as neuropsychological contributors to impulsivity as a personality dimension. Nevertheless, despite some studies reporting associations between impulsive traits and behavior-based measures of these so-called executive functions ([Keilp et al., 2005](#); [Pietrzak et al., 2008](#)), others have found no such relationship ([Horn et al., 2003](#); [Reynolds et al., 2006](#)). These mixed findings may stem from the multi-faceted nature of impulsivity, an issue that has not received much attention in research on cognition and personality until recently.

Although several measures of trait impulsivity exist, arguably the one with the longest track record that is still in wide use is the Barratt Impulsiveness Scale (BIS; [Stanford et al., 2009](#)). The most recent version, the BIS-11, measures three related but dissociable sub-dimensions of impulsivity. While some studies have examined zero-

order correlations between BIS-11 facets and executive function measures ([Enticott et al., 2006](#); [Gorlyn et al., 2005](#); [Keilp et al., 2005](#); [Pietrzak et al., 2008](#); [Spinella, 2004](#); [Swann et al., 2002](#)), to our knowledge no work has examined whether these facets have unique relationships with measures tapping executive functions. The BIS-11 subscales have specific correlates with clinically important uncontrolled behaviors such as binge drinking ([Carlson et al., 2010](#)) and manic symptoms ([Swann et al., 2008](#)). As such, the relationships between unique variance in one facet and measures of executive function may provide clues to the cognitive processes underlying personality risk for disinhibited behaviors.

Further, most studies examining BIS-11 subscales have used behavior-based measures of executive functions (see [Harmon-Jones et al., 1997](#); [Goya-Maldonado et al., 2010](#) for exceptions). Some individuals with high trait impulsivity may compensate for their cognitive limitations by engaging other resources with little observable difference in behavior. Therefore, behavioral responses may be less sensitive indices of underlying executive limitations than are direct measures of neural activity (e.g., [Goya-Maldonado et al., 2010](#)).

Event-Related Potentials (ERPs) are measures of neural activity that have been used in studies of executive functions (e.g., [Fallgatter et al., 1997](#); [Roberts et al., 1994](#); [Bokura et al., 2001](#); [Bruin and Wijers, 2002](#); [Kirmizi-Alsan et al., 2006](#); [Smith et al., 2008](#); [Tekok-Kilic et al., 2001](#)). Some ERP measures have been related to trait impulsivity in tasks that are influenced by a variety of demands, not all of which are necessarily directly related to executive functions (e.g.,

* Corresponding author at: Department of Psychology, University of Minnesota Duluth, 320 Bohannon Hall, 1207 Ordean Court, Duluth, MN 55812–3010, USA. Tel.: +1 218 726 6428; fax: +1 218 726 7186.

E-mail address: carls116@umn.edu (S.R. Carlson).

Chen et al., 2007; De Pascalis et al., 2004; Harmon-Jones et al., 1997; Justus et al., 2001; Moeller et al., 2004; Russo et al., 2008; Venables et al., 2011). The present study attempts to clarify the relationship between facets of impulsivity and executive processes by examining partial relationships between BIS-11 subscales and both behavior and ERP-based measures.

1.1. Trait impulsivity and the BIS-11

Conceptualized as “a predisposition toward rapid, unplanned reactions to internal or external stimuli without regard to the negative consequences of these reactions” (Moeller et al., 2001), impulsivity is a multidimensional personality trait present in both normal and clinical populations. Although impulsivity has often been explored as a unitary trait, increasing attention has been paid to the multiple facets of impulsivity (Gay et al., 2008; Pietrzak et al., 2008; Whiteside and Lynam, 2001).

The BIS-11 (Patton et al., 1995) taps three such sub-traits. The first facet, Attentional Impulsiveness (AI), is characterized by difficulty in concentrating. Longer drinking sessions have been associated more strongly with this sub-trait than the other two sub-traits (Balodis et al., 2009). Non-Planning Impulsiveness (NPI), the second sub-trait, is characterized by a lack of impulse control and planning for the future. Only this sub-trait has been significantly associated with a greater number of daily cigarettes smoked among alcoholic subjects (Dom et al., 2006). Further, both of these above sub-traits have been associated with physical aggression (Harmon-Jones et al., 1997). Characterized by acting without thinking, the third sub-trait, Motor Impulsiveness (MI), appears to be more strongly related to binge drinking than the other facets (Carlson et al., 2010). The report of dissociations between the relationships among these three facets and behavioral problems suggests a relative independence of the three impulsivity sub-traits. These facets may be dissociable, in part, because they are differentially related to cognitive processes subserving executive functions.

1.2. Executive functions and the BIS-11

Impulsivity may reflect individual differences in cognitive function underlying the ability to exercise self-control. Characterized as a set of higher order cognitive functions, executive functions have been conceptualized as related but distinct processes (Baddeley, 1996; Miyake et al., 2000) necessary for planning, executing, and monitoring purposeful behavior. The relationships between BIS-11 scales and executive functions have not been consistently reported (e.g., Horn et al., 2003; Reynolds et al., 2006). Nevertheless, it appears that MI is associated with several behavioral measures of executive disinhibition, usually with somewhat higher magnitude correlations than the other BIS-11 subscales (although see Enticott et al., 2006 for an exception¹). This is the case on tasks such as the Go/No-Go (e.g., $r=0.35$ vs. $r=0.21$ – 0.26 for other scales, Spinella, 2004; $r=0.34$ vs. $r=0.13$ – 0.28 , Keilp et al., 2005), antisaccade ($r=0.50$ vs. $r=0.26$ – 0.43 ; Spinella, 2004), Stop Signal ($r=0.34$ vs. $r=0.03$ – 0.06 , Gorlyn et al., 2005) and the Trail Making B ($r=0.29$ vs. $r=-0.06$ – 0.26 ; Keilp et al., 2005) Tasks². On the other hand, longer completion time for tasks as a whole or on individual trials (i.e., reaction time) has been associated with NPI ($r=0.34$ vs. $r=0.16$ – 0.27 ,

Gorlyn et al., 2005; $r=0.58$ vs. $r=.18$ – 0.23 , Pietrzak et al., 2008). As with the zero-order correlations between MI and measures of disinhibition, these correlations have been at least marginally and sometimes substantially greater than those observed with the other subscales.

To date, however, no studies have used partial correlation, multiple regression, or similar techniques to examine whether or not a relationship between a BIS-11 facet and an executive function measure identified by simple zero-order correlations is maintained after removing the variance shared between that subscale and the others. This is important as the BIS-11 subscales are intercorrelated ($0.39 \leq r \leq 0.50$, Stanford et al., 2009), but have some unique correlates (e.g., Carlson et al., 2010; Swann et al., 2008). Many studies that identify a zero-order correlation between one executive function task and a subscale, report that other BIS-11 facets are also significantly correlated with that cognitive measure (e.g., Enticott et al., 2006; Keilp et al., 2005; Pietrzak et al., 2008; Spinella, 2004; Swann et al., 2002). The identification of relationships between BIS-11 facets and executive functions, while controlling for variance in other subscales, would support the hypothesis that BIS-11 facets are differentially related to cognitive control mechanisms.

1.3. Executive functions and ERPs

Two ERP measures used to examine cognitive processes core to many executive functions are the amplitudes of the so called No-Go P3 (e.g., Pfefferbaum et al., 1985; Fallgatter et al., 1999) and N2 (e.g., Falkenstein, 2006) components. The No-Go P3 has been implicated in inhibiting a prepotent response during a variation of a continuous performance task (CPT) called the AX-CPT (Fallgatter et al., 1997; Kirmizi-Alsan et al., 2006; Roberts et al., 1994) and during Go/No-Go tasks (Bokura et al., 2001; Bruin and Wijers, 2002; Kirmizi-Alsan et al., 2006; Smith et al., 2008; Tekok-Kilic et al., 2001). Although commonly viewed as an index of inhibition, per se, some have suggested No-Go P3 amplitude may reflect a post-response evaluation of successful inhibition (Roche et al., 2005; Schmajuk et al., 2006). This frontocentral maximal No-Go P3 is contrasted with a parietal maximal P3, often called P3b, which appears to index top-down attention, as well as working memory updating (Polich, 2007).

The No-Go N2 has a frontocentral maximum and is typically larger in association with No-Go compared to Go stimuli. It appears to potentially reflect a number of processes involved in cognitive control of behavior (as reviewed by Folstein and van Petten, 2008) including pre-motor inhibition putatively related to right orbitofrontal functioning and conflict detection associated with the anterior cingulate cortex when there are multiple competing response options (Falkenstein, 2006). It is in contrast to a more posterior family of N2 amplitudes involved in stimulus mismatch detection (Folstein and van Petten). Thus, the No-Go P3 and N2 represent prime targets to further expand our understanding of the relationship between impulsivity and executive functions necessary for behavioral control under conditions of conflict.

Although the relationship between reduced P3 amplitude and impulsivity appears to be robust (Chen et al., 2007; De Pascalis et al., 2004; Russo et al., 2008), most researchers used a general measure of impulsivity rather than examining the sub-traits. One exception is the work of Harmon-Jones et al. (1997). In their study, AI and NPI were associated with reduced parietal P3 amplitude more indicative of top-down attention than inhibition (e.g., Polich, 2007) in a traditional CPT that required more sustained attention than an AX-CPT (Smid et al., 2006). MI, however, was not related to P3. Further, none of the BIS-11 facets in this study were significantly associated with P3 in an oddball task not thought to tap inhibition directly. Although this study suggests possible differential relationships among the impulsivity facets and an ERP component, it is limited in its ability

¹ Enticott et al., 2006, however, reported that Pearson correlation coefficients between MI and inhibitory control variables ranged from $0.13 \leq r \leq 0.38$, whereas correlations ranged from $.05 \leq r \leq 0.56$ for NPI.

² Keilp et al. (2005) reported correlations between BIS-11 facets and executive function measures controlling for variance associated with age and education, but did not report partial relationships between an impulsivity facet and an executive function measure controlling for the other two subscales.

to identify unique relationships as it involved a small sample and did not evaluate partial relationships.

1.4. The current study

The present study investigates the relationship between facets of the BIS-11 and behavioral and electrophysiological measures of cognitive control and inhibition. We administered the BIS-11 to undergraduate students who also completed an expectancy AX-CPT (MacDonald and Carter, 2003). Emerging adulthood is an important period in which to study impulsivity. During this life stage there is ongoing maturation of the neural systems underlying executive processes (Best and Miller, 2010; Hwang et al., 2010; Qiu et al., 2010; Sowell et al., 2001). The variability in this maturational process is evident in the differing developmental trajectories of impulsivity and alcohol use across this age range (Littlefield et al., 2010). Furthermore, risky behaviors related to impulsivity, like substance use, tend to reach their maximum frequency during this period (Arnett, 2000; Arbour-Nicotopoulos et al., 2010).

We chose an AX-CPT for two reasons. First, it has been used to examine conflict detection among competing response options and inhibition of a dominant response in previous ERP studies (Cohen et al., 1999; Rush et al., 2006). Second, although multiple task conditions are likely to require response inhibition and conflict detection associated with frontocentral P3 and N2 amplitude, the task involves conditions that should not engage inhibitory processes to the same extent and therefore can serve as control conditions. Given that zero-order correlations between MI and measures of disinhibition appear to be greater than those involving other BIS-11 facets and evidence suggesting the anterior No-Go P3 indexes response inhibition, we hypothesize a unique negative correlation between MI and anterior P3 amplitude in a task condition requiring inhibition of a prepotent response. We also predict reduced frontal N2 (i.e., less negative amplitude) to be associated with MI, as the frontocentral N2 amplitude may index conflict monitoring necessary for response inhibition, pre-motor inhibition itself, or both. Further, given that NPI has been associated more often with delayed reaction time (RT; Gorlyn et al., 2005; Pietrzak et al., 2008), we anticipate a specific relationship between this facet and RT on the task.

2. Methods

2.1. Subjects

The subjects were 85 right-handed undergraduate students between the ages of 17.56 and 22.35 ($M = 19.95$; $SD = 1.14$) years, who were enrolled at the University of British Columbia. Twenty-eight (32.9%) were male. Age did not significantly differ by sex, $t(83) = 0.08$, $p = 0.934$. More than half of our sample identified themselves as being East Asian or of East Asian descent (64.7%), while 29.4% identified as being European or of European descent, 7.1% as being Indian-South Asian or of Indian-South Asian descent, and 2.4% identified as Middle Eastern or of Middle Eastern descent. Participants were allowed to identify as many ethnic groups as they wished and 3.5% self-identified as bi-ethnic, so the aforementioned percentages do not sum to 100%. There was not a significant difference by sex in the proportion of East Asian (Fisher's Exact Test $p = 0.056$), European (Fisher's Exact Test $p = 0.801$), or other (Fisher's Exact Test $p = 0.108$) ethnic ancestry groups. Subjects must have been free of neurological problems, have normal or corrected-to-normal vision, and have been speaking English as their primary language for a minimum of two years. Handedness was assessed by the Edinburgh Handedness Inventory (Oldfield and Edinburgh, 1971). Participants were recruited through an online human subject pool system coordinated by the university's psychology department, and were compensated with credits for a psychology course. They gave written, informed

consent prior to participation. The institutional research ethics board approved all procedures in this study.

2.2. Trait impulsivity

We measured trait impulsivity with the BIS-11 (Patton et al., 1995). This is a 30 item questionnaire that, as previously mentioned, taps three sub-traits: Attention (AI), Motor (MI) and Non-Planning (NPI) Impulsiveness. All items are answered using a 4-point scale ranging from 1 to 4. Patton et al. (1995) reported significant correlations among these subscale scores (ranging from 0.15 to 0.42). In the present sample Cronbach's alpha was 0.80 for the total score; 0.69 for AI; 0.60 for MI; and 0.64 for NPI. These internal consistencies were similar to those reported in a recent review of the BIS-11 (Stanford et al., 2009; alpha ranged from .59 to .83). Stanford et al. (2009) also reported moderate test-retest reliability for the BIS total and subscale scores over an interval of one month (reliabilities ranged from 0.61 to 0.83). The BIS-11 has been developed with and extensively administered to college student samples and displays an array of relevant correlates supporting the validity of its scores in undergraduates (Patton et al., 1995; Stanford et al., 1996, 2009).

2.3. AX-CPT

The Expectancy "AX" continuous performance task (AX-CPT; MacDonald and Carter, 2003) was used to assess cognitive control. In this task, pairs of letters were presented, one letter after the other. The letter-pairs were presented on a computer screen 110 cm away from participants sitting in a darkened room. The first letter, either A or B, served as a cue. The second letter, either X or any other letter except for A, B, or K (to avoid confusion with X), which will be referred to as Y hereafter, was considered the probe. Thus, there were four types of trials, that is, A–X, B–X, A–Y, and B–Y. Participants were instructed to press a target button with their thumb only when A was followed by X. If it was any other combination of letters (A–Y, B–X, and B–Y), they were required to press another button with the other thumb. The target and non-target buttons were the left- and right-most buttons on a Psychology Software Tools, Inc. Deluxe Serial Response Box with the assignment of right or left buttons to target condition counterbalanced across subjects. Five blocks of 70 trials each were presented. Within each block, the most common trial, the A–X sequence, was present 70% of the time, with the remaining 30% equally distributed among the other three sequences. Order of presentation of these pairs was random within blocks. The A–Y condition requires inhibition of a prepotent response, as the subject is primed to make an incorrect target response by the cue. On the other hand, the B–X condition requires subjects to maintain a representation in working memory of the context provided by the cue stimulus in order to press the correct button. This condition also is the only condition in which a non-target response is made to the X-probe that is most often associated with the target response.

Stimulus presentation and behavioral response acquisition were controlled by E-Prime Version 1.1 software. Letters were presented in white against a black background in 72-point bold Arial black font. The visual arc subtended by the stimuli ranged from 1.77° to 2.14°. Stimulus duration was 150 ms, and the ISI was 4000 ms. The ITI between the off-set of the probe and onset of the next cue was 950 ms. Subjects were given 10 practice trials, with the same proportions of each condition as the main task, to ensure they understood the instructions. Subjects who did not achieve at least 90% accuracy on the practice had to complete another block of practice trials before proceeding to the actual task. Subjects were told to respond as quickly and accurately as possible.

2.4. Psychophysiological recording

Electroencephalographic (EEG) data were acquired with a Brain Products Inc, QuickAmp 72 System. Brain Vision Recorder and Brain Vision Analyzer software were used to record the data, and to process and score ERPs, respectively. Brain activity was recorded from the F3, Fz, F4, C3, Cz, C4, P3, Pz and P4 scalp sites using Ag/AgCl electrodes in an electrode cap. The ground electrode was placed at the AFz site. Recordings were referenced offline to averaged mastoids. Impedance was kept under 5 k Ω . Electrooculographic (EOG) data were also recorded with Ag/AgCl electrodes, placed above and below the right eye to detect blinks, and placed to the side of each eye to detect horizontal eye movement. Impedance for EOGs was kept under 10 k Ω . Both EEGs and EOGs were sampled at 500 Hz continuously. Signals were digitally filtered offline with high and low pass frequency filters at 0.01 and 30 Hz (12 dB/octave roll-off), respectively. The algorithm of Gratton et al. (1983) was used to correct for the effects of blinks and ocular artifacts on EEG.

Signals were segmented into 1500 ms epochs including a 500 ms baseline prior to probe stimulus onset. Trial epochs were baseline-corrected prior to signal averaging. Only trials associated with accurate behavioral responses were included in data analyses, and these trials were averaged separately for each condition. P3 was identified initially by an interactive computer algorithm as the most positive voltage between 200 and 800 ms following stimulus onset. Similarly, N2 was identified with the same algorithm, but as the most negative voltage in a window from 150 to 540 ms following stimulus onset. These time windows for component selection were based on grand averages derived from this sample. They were deliberately broad in order not to miss a component and are consistent with other studies (Jonkman, 2006; Jonkman et al., 2003; Overtom et al., 1998). These selections were manually verified by a rater who compared the waveforms across task conditions and all nine scalp sites. The rater corrected cases in which components were misidentified. Amplitude was defined as the difference in voltage between the mean baseline and the identified peak (for P3), or “valley” (for N2). Latency was the time from stimulus onset to the peak of the P3 or N2 component. To reduce the possibility that differences in P3 and N2 amplitudes were because of variations in task compliance, those who responded with more than 20% error in the A–X (i.e., the prepotent response priming condition) or A–Y (i.e., the condition requiring inhibition of a primed prepotent response) trials were excluded from analysis. In addition to the 85 subjects included in analyses, 9 participants completed the task but were excluded because of poor task performance. An additional 27 subjects participated partially in that they completed the questionnaires but were excluded because of computer failures or data recording problems (e.g. excessive scalp impedance), or for not meeting study exclusion criteria related to ERP recordings.

2.5. Assessment of potential substance use confounds

Substance use is significantly associated with BIS-11 scale scores (Stanford et al., 2009) and P3 amplitude (e.g., Iacono et al., 2008). We thus asked several questions following the format recommended by the National Institute on Alcohol Abuse and Alcoholism (NIAAA, 2003) for assessing drinking behavior in research. This included the recommended questions regarding frequency of alcohol consumption, the average quantity subjects consumed when they drank, maximum lifetime number of drinks consumed within 24 h and the frequency of binge drinking in the past year. Quantity $\times$ frequency estimates of alcohol consumption in the past year were produced by multiplying the answers to the first two questions. A binge was defined as having a certain number of drinks or more within a two hour period. This number was 4 or more for women and 5 or more for men (NIAAA, 2004). In addition, subjects were also asked what their age was when they first tried alcohol without their parents'

permission. As some subjects had not used alcohol at any point in their lives, they could not have an age at first use. For this reason, we analyzed “years since first use” instead. This was defined as the subjects' age at first use subtracted from their current age. Subjects who never used the substance were assigned a value of 0 years since first use. A similar question was asked regarding the age at which they were first intoxicated and was scored in a parallel way as years since first intoxication. For all questions a drink was defined as “half an ounce (30 ml) of absolute alcohol (e.g. a 12 oz/355 ml can or glass of beer or cooler, a 5 oz/150 ml glass of wine, or a drink containing 1 shot of liquor)” as in the NIAAA recommendations.

We also asked about the use of illicit drugs. Our drug use measures were adapted from question 13 of the Adolescent Drug Involvement Scale (ADIS; Moberg and Hahn, 1991). This question asked participants to rate their frequency of drug use on an 8-point scale ranging from “never used” to “several times a day”. We adapted the question to enquire about the following substances: marijuana or hashish; LSD, psilocybin, peyote, other hallucinogens (acid); amphetamines; cocaine; barbiturates; PCP; heroin; other opiates; Valium and other tranquilizers; and ecstasy. Given the relatively greater prevalence of cannabis use we considered the respondents' rating for this class of substance separately. For all other drugs, we summed the ratings across substances to give a “hard drug” use measure. As the alcohol and drug use measures, with the exception of years since first drink and years since first intoxication, were positively skewed we applied a natural log transformation after adding a constant of 1.

In order to simplify analyses and reduce multi-collinearity among substance use control variables, a principal component analysis (PCA) was conducted. Only one component was extracted as it was the only one with an eigenvalue > 1 (i.e., 3.93). It accounted for 56.18% of the variance across measures. The log-transformed quantity $\times$ frequency measure of past year drinking had the highest loading (0.938). The component loading for maximum consumption was 0.888, binge frequency was 0.861, cannabis use was 0.705, age since first intoxication was 0.648, age since first drink was 0.623, and hard drug use loaded 0.485. This component will be referred to as substance use hereafter.

3. Results

3.1. Descriptive statistics

Descriptive statistics for BIS-11 and substance use measures are provided on the diagonal of Table 1. A potential concern when conducting research on impulsivity in community or college samples is a possible restriction of range caused by individual differences among subjects not extending into the severe end of the continuum. Using the guidelines of Stanford et al. (2009), however, 17.6% of our sample would be classified as highly impulsive (BIS-Total $\geq$ 72), while 8.2% were in the low impulsive range (BIS-Total $\leq$ 51). Guidelines for clinically significant impulsivity do not, to our knowledge, exist for the subscales. However, the minimum and maximum values were 10 and 26 out of possible values ranging from 8 to 32 for AI, 15 and 31 out of a possible range from 11 to 44 for MI and 15 and 39 out of a possible range of 11–44 for NPI. The percentage of the entire possible range that was observed for each scale was 67% for AI, 48% for MI and 73% NPI. The majority of the scales were represented for AI and NPI, but not MI. As a result, any relationships identified between MI and measures from the AX-CPT are likely conservative underestimates due to restriction of range. Furthermore, the relatively low internal consistencies we reported in the method section also suggest that any relationships we observe are likely conservative underestimates.

Zero-order correlations among these variables are also provided in Table 1. Substance measures tended to be moderately to highly inter-correlated. BIS-11 scales tended to be moderately correlated with each other. With a few exceptions, expected significant correlations

Table 1

Correlations between BIS-11 scales and substance use measures with means (standard deviations) on the diagonal.

	BIS-11 total	AI	MI	NPI	Binge Freq.	Alc. Max.	Alc. Q×F	Age at 1st drink	Age at 1st intox.	Cannabis	Hard drug	Substance use
BIS-11 total	63.04 (9.29)											
Attentional Impulsiveness (AI)	0.77***	16.91 (3.51)										
Motor Impulsiveness (MI)	0.77***	0.36***	22.21 (3.87)									
Non-Planning Impulsiveness (NPI)	0.85***	0.53***	0.46***	23.92 (4.24)								
Binge Freq.	0.22*	0.19~	0.16	0.19~	9.72 (18.94)							
Alc. Max.	0.16	0.15	0.15	0.09	0.78***	8.62 (7.77)						
Alc. Q×F	0.20~	0.20~	0.17	0.11	0.85***	0.91***	124.50 (243.78)					
Age at 1st drink	0.14	0.15	0.04	0.13	0.44***	0.42***	0.44***	3.70 (2.56)				
Age at 1st intox.	0.06	0.11	−0.04	0.06	0.41***	0.45***	0.45***	0.42***	2.00 (2.34)			
Cannabis	0.14	−0.01	0.19~	0.14	0.50***	0.53***	0.54***	0.30**	0.19	0.58 (1.00)		
Hard drug	0.12	0.01	0.13	0.13	0.20~	0.23*	0.27*	0.08	0.20~	0.46***	0.25 (0.89)	
Substance use	0.20~	0.17	0.16	0.16	0.86***	0.90***	0.92***	0.60***	0.60***	0.69***	0.42***	−0.02 (0.95)

Note: Correlations involving substance use measures used the natural logarithm of the measure + 1. Means and standard deviations are based on the untransformed data. Binge Freq. = frequency of alcohol binges in past year (in days); Alc. Max. = maximum number of drinks consumed in 24 hours (in number of standardized drinks); Alc. Q×F = past year drinking quantity per occasion multiple by number of drinking occasions (in number of standardized drinks); Cannabis = estimated cannabis use; substance use is the first principal component from a principal component analysis of the substance measures.

N = 85.

~0.05 < p ≤ 0.10; *0.01 < p ≤ 0.05; **0.001 < p ≤ 0.01; *** p ≤ 0.001.

between BIS-11 scales and substance use measures were not observed. Table 2 provides descriptive statistics for P3 and N2 amplitude and latency, as well as RT and response accuracy for all conditions in the AX-CPT.

No substance use measure differed significantly between the sexes (all p -values ≥ 0.298). Of the BIS-11 measures, only MI differed significantly between men ($M = 21.00$, $SD = 4.23$) and women ($M = 22.81$, $SD = 3.58$), $t(83) = 2.06$, $p = 0.043$. All subsequent analyses involved sex-corrected BIS-11 subscale and substance use scores, produced by saving the unstandardized residuals from linear regressions predicting each measure from dummy coded sex. This was done in order to reduce the chance that relationships between MI and AX-CPT measures are spurious byproducts of sex differences in the trait. For consistency's sake we also sex-corrected the other personality trait and substance use measures. We also evaluated differences between men and women in zero-order correlations between ERP or behavioral measures and substance use and each BIS-11 facet. There were no significant differences in correlation between men and women (all p -values ≥ 0.085).³

3.2. P3 amplitude

In order to evaluate the effects of the BIS-11 subscales on P3 amplitude, we conducted mixed models analyses using SPSS 18.0 with

³ In order to evaluate if the relatively high proportions of subjects of East Asian and European background affected our findings we compared East Asian subjects to all others in terms of substance use and BIS-11 facet means. We also compared subjects of European-ancestry to all others on the sex-corrected versions of these variables. Only substance use differed significantly [East Asian ($N = 55$; $M = -0.27$; $SD = 0.78$) vs. Others ($N = 30$; $M = 0.50$; $SD = 1.03$), $t(83) = 3.896$, $p < 0.001$; European ($N = 25$; $M = 0.59$; $SD = 0.91$) vs. Others ($N = 60$; $M = -0.25$; $SD = 0.86$), $t(83) = 4.035$, $p < 0.001$]. We also evaluated differences between East-Asian and European subjects in partial correlations between ERP or behavioral measures and substance use and each BIS-11 facet correcting for sex. There were no significant differences in correlation between these groups of subjects (all p -values ≥ 0.116).

maximum-likelihood estimation and unstructured residual covariance. Electrode site (Fz, Cz, and Pz) and task condition (A–X, A–Y, B–X, and B–Y) were repeated within-participants factors with data nested within the individual participant. Sex (male, female) was evaluated as a between-participant fixed effect with substance use, AI, MI, and NPI as between-participant continuous measures. All 2-way interactions involving the between-subjects variables interacting with Condition and Site were included in the model. All 3-way interactions involving between-subjects variables and Condition×Site were also evaluated. Significant main effects for categorical variables were evaluated with post hoc tests of group differences. In order to do this, sub-models were fit comparing just two levels of a factor at a time with all other variables included in the model. Bonferroni corrections were applied to the significance levels such that α for comparisons across conditions was 0.008 (0.05/6 pair-wise comparisons across 4 conditions) and 0.017 across scalp-sites (0.05/3 pair-wise comparisons across 3 sites). For Sex×Condition interactions, sex differences were compared within conditions ($\alpha = 0.05/4$ comparisons across 4 conditions = 0.0125). For Sex×Site interactions α was 0.05/3 sex comparisons at 3 sites or 0.0167. Standardized partial regression coefficients were estimated from the model for relationships between P3 amplitude and sex-corrected continuous measures. Results for the F -tests within the model are presented in Table 3 with estimated regression coefficients presented in Table 4. Significant effects will be described in what follows.

There was a significant main effect for Condition. Specifically, P3 in the A–Y condition was significantly greater than in the A–X, B–X and B–Y conditions (all p -values < 0.001). Further, amplitude in the B–Y condition was significantly greater than that in the A–X condition, $F(1,85) = 16.15$, $p < 0.001$. P3 amplitude also differed significantly across scalp sites ($Fz < Cz < Pz$, p -values < 0.001). Condition and site main effects must be interpreted in the context of a significant Site×Condition interaction reflecting a trend for A–Y amplitude to be relatively larger at the Fz site than at Cz or Pz. Women had significantly greater P3 amplitude than men, but Condition interacted

Table 2

Means and standard deviations by participant sex for P3 amplitude, P3 latency, N2 amplitude, N2 latency, reaction time, and accuracy.

Measure		All Participants (n = 85)		Male (n = 28)		Female (n = 57)		Sex-Difference	
		Mean	(SD)	Mean	(SD)	Mean	(SD)	t (83)	p
P3 amplitude (μV)									
Fz	A–X	5.21	(5.03)	4.65	(4.81)	5.48	(5.16)	0.72	0.475
	A–Y	11.20	(6.92)	9.09	(5.00)	12.24	(7.51)	2.00	0.048
	B–X	6.58	(5.29)	5.81	(4.59)	6.97	(5.60)	0.95	0.345
	B–Y	6.85	(5.44)	5.76	(4.88)	7.38	(5.66)	0.14	0.198
Cz	A–X	8.66	(5.15)	7.65	(4.89)	9.16	(5.25)	1.27	0.208
	A–Y	13.58	(5.99)	11.07	(4.47)	14.81	(6.29)	2.81	0.006
	B–X	9.67	(5.46)	9.28	(4.86)	9.87	(5.77)	0.47	0.643
	B–Y	10.39	(5.73)	9.08	(4.95)	11.03	(6.02)	1.48	0.142
Pz	A–X	12.26	(4.07)	10.83	(3.37)	12.96	(4.23)	2.33	0.022
	A–Y	16.58	(5.37)	13.94	(3.76)	17.88	(5.59)	3.37	0.001
	B–X	11.50	(4.48)	10.04	(3.63)	12.22	(4.71)	2.16	0.034
	B–Y	13.40	(4.92)	11.22	(4.09)	14.47	(4.97)	3.00	0.004
P3 latency (ms)									
Fz	A–X	531.20	(59.40)	530.93	(59.98)	531.33	(59.65)	0.03	0.977
	A–Y	526.66	(49.00)	523.21	(55.96)	528.35	(45.63)	0.45	0.652
	B–X	623.65	(89.05)	625.21	(94.98)	622.88	(86.86)	0.11	0.910
	B–Y	609.93	(104.32)	618.93	(104.32)	605.51	(80.95)	0.65	0.516
Cz	A–X	551.84	(61.13)	540.36	(65.30)	557.47	(58.74)	1.22	0.227
	A–Y	545.58	(52.25)	533.79	(56.26)	551.37	(49.65)	1.47	0.146
	B–X	641.13	(82.12)	649.79	(81.68)	636.88	(82.72)	0.68	0.499
	B–Y	627.76	(75.30)	649.14	(90.27)	617.26	(65.09)	1.86	0.066
Pz	A–X	537.46	(59.91)	535.07	(61.33)	538.63	(59.71)	0.26	0.799
	A–Y	577.27	(49.41)	573.36	(53.44)	579.19	(47.68)	0.51	0.612
	B–X	620.45	(95.71)	630.64	(101.07)	615.44	(93.48)	0.69	0.495
	B–Y	613.65	(75.58)	618.86	(85.65)	611.09	(70.80)	0.44	0.659
N2 Amplitude (μV)									
Fz	A–X	−3.50	(4.11)	−4.15	(3.47)	−3.18	(4.39)	1.02	0.309
	A–Y	−3.97	(6.13)	−4.97	(5.07)	−3.48	(6.57)	1.06	0.294
	B–X	−5.31	(5.21)	−4.61	(4.02)	−5.65	(5.70)	0.87	0.387
	B–Y	−5.16	(5.11)	−5.19	(4.04)	−5.15	(5.60)	0.03	0.977
Cz	A–X	−1.03	(4.22)	−2.32	(3.65)	−0.40	(4.37)	2.01	0.048
	A–Y	−2.04	(5.45)	−3.58	(5.18)	−1.29	(5.47)	1.84	0.069
	B–X	−3.47	(5.19)	−2.99	(4.05)	−3.71	(5.68)	0.60	0.548
	B–Y	−3.43	(5.13)	−3.53	(4.35)	−3.38	(5.51)	0.12	0.902
Pz	A–X	3.41	(3.56)	2.35	(2.62)	3.93	(3.86)	1.95	0.054
	A–Y	2.00	(4.90)	1.00	(4.21)	2.50	(5.17)	1.33	0.187
	B–X	2.26	(4.12)	1.91	(3.34)	2.43	(4.46)	0.55	0.585
	B–Y	2.01	(4.63)	1.78	(3.59)	2.12	(5.09)	0.32	0.750
N2 latency (ms)									
Fz	A–X	400.02	(30.88)	391.71	(35.52)	404.11	(27.77)	1.76	0.082
	A–Y	395.93	(26.98)	397.93	(26.87)	394.95	(27.22)	0.48	0.635
	B–X	402.99	(29.60)	400.93	(29.93)	404.00	(29.65)	0.45	0.656
	B–Y	402.89	(25.50)	396.79	(28.91)	405.89	(23.33)	1.56	0.122
Cz	A–X	391.25	(37.45)	384.64	(38.84)	394.49	(36.67)	1.14	0.257
	A–Y	393.81	(28.39)	391.00	(29.64)	395.19	(27.91)	0.64	0.525
	B–X	398.49	(32.71)	391.79	(36.79)	401.79	(30.31)	1.33	0.187
	B–Y	404.35	(24.43)	395.57	(30.23)	408.67	(19.94)	2.39	0.019
Pz	A–X	358.64	(39.35)	356.71	(38.12)	359.58	(40.24)	0.31	0.754
	A–Y	381.51	(36.12)	382.43	(34.61)	381.05	(37.13)	0.16	0.870
	B–X	381.60	(39.89)	377.14	(38.90)	383.79	(40.52)	0.72	0.474
	B–Y	392.82	(34.92)	380.29	(38.04)	398.98	(31.84)	2.38	0.019
Reaction time (ms)									
	A–X	462.49	(72.37)	455.76	(63.52)	465.80	(76.66)	0.60	0.551
	A–Y	604.10	(83.79)	600.93	(80.39)	605.66	(86.07)	0.24	0.808
	B–X	438.34	(114.12)	431.97	(121.09)	441.47	(111.52)	0.36	0.720
	B–Y	440.36	(107.49)	442.87	(122.17)	439.13	(100.65)	0.15	0.881
Accuracy (% correct)									
	A–X	95.76	(4.26)	97.74	(2.26)	94.79	(4.68)	3.47	0.001
	A–Y	88.34	(8.72)	91.02	(6.06)	87.02	(9.54)	2.05	0.044
	B–X	92.87	(7.72)	94.69	(6.63)	91.98	(8.11)	1.27	0.207
	B–Y	95.66	(10.71)	95.20	(17.14)	95.89	(5.48)	0.81	0.418

Note: P3, N2, reaction time and accuracy are reported from trials with correct behavioral responses to probe stimuli. Significant differences between men and women are in bold font.

significantly with sex. That is, women had significantly greater amplitude than men only in the A–Y [$F(1,85) = 21.97$, $p = 0.002$] condition.

There was a significant main effect for MI to be inversely related to P3 amplitude across task conditions and scalp sites, reflected by an average standardized partial regression coefficient across sites and

Table 3
Main and interaction fixed effects for mixed models evaluating the relationships between substance use and BIS-11 facets with P3 amplitude, N2 amplitude, reaction time, and response accuracy.

	P3 amplitude			N2 amplitude			RT			Response accuracy		
	F	df	P	F	df	P	F	df	p	F	df	P
Cond.	41.69	3,85	<0.001	17.47	3,85	<0.001	252.64	3,85	<0.001	45.03	3,85	<0.001
Site	68.43	2,85	<0.001	129.57	2,85	<0.001	–	–	–	–	–	–
Cond. × Site	8.71	6,85	<0.001	5.91	6,85	<0.001	–	–	–	–	–	–
Sex	5.61	1,85	0.020	0.80	1,85	0.373	0.07	1,85	0.793	5.43	1,85	0.022
Sex × Cond.	2.85	3,85	0.042	4.35	3,85	0.007	0.78	3,85	0.508	0.70	3,85	0.556
Sex × Site	0.80	2,85	0.451	0.65	2,85	0.526	–	–	–	–	–	–
Sex × Cond. × Site	1.54	6,85	0.175	2.56	6,85	0.025	–	–	–	–	–	–
MI	5.47	1,85	0.022	5.74	1,85	0.019	0.08	1,85	0.776	0.95	1,85	0.334
MI × Cond.	0.36	3,85	0.784	3.33	3,85	0.023	0.32	3,85	0.809	1.55	3,85	0.208
MI × Site	0.84	2,85	0.437	1.53	2,85	0.223	–	–	–	–	–	–
MI × Cond. × Site	0.93	6,85	0.477	2.25	6,85	0.046	–	–	–	–	–	–
NPI	1.41	1,85	0.238	1.89	1,85	0.173	1.58	1,85	0.212	0.87	1,85	0.354
NPI × Cond.	0.72	3,85	0.541	0.26	3,85	0.856	2.61	3,85	0.057	1.49	3,85	0.225
NPI × Site	2.22	2,85	0.115	0.16	2,85	0.854	–	–	–	–	–	–
NPI × Cond. × Site	3.89	6,85	0.002	2.56	6,85	0.025	–	–	–	–	–	–
AI	0.05	1,85	0.826	0.01	1,85	0.927	0.43	1,85	0.512	0.09	1,85	0.767
AI × Cond.	0.46	3,85	0.709	8.56	3,85	<0.001	1.39	3,85	.250	1.04	3,85	0.381
AI × Site	0.40	2,85	0.674	0.56	2,85	0.575	–	–	–	–	–	–
AI × Cond. × Site	0.46	6,85	0.834	2.31	6,85	0.041	–	–	–	–	–	–
Sub.	0.23	1,85	0.633	2.96	1,85	0.089	1.21	1,85	0.275	6.88	1,85	0.010
Sub. × Cond.	2.35	3,85	0.079	0.47	3,85	0.701	2.69	3,85	0.051	3.03	3,85	0.034
Sub. × Site	0.16	2,85	0.855	2.49	2,85	0.089	–	–	–	–	–	–
Sub. × Cond. × Site	0.76	6,85	0.602	1.01	6,85	0.426	–	–	–	–	–	–

Note: MI = Barratt Impulsiveness Scale 11 (BIS-11) Motor Impulsiveness; NPI = BIS-11 Non-Planning Impulsiveness; AI = BIS-11 Attentional Impulsiveness; Sub. = substance use is the first principal component from a principal component analysis of substance use measures. All continuous predictor variables were sex-corrected and were the unstandardized residuals from regressions using dummy coded sex as a predictor. Sex is dummy coded (0 = male, 1 = female). Cond. = task condition (A–X, A–Y, B–X, B–Y). Site = scalp site (Fz, Pz, Cz). RT = reaction time. Statistically significant effects are in bold font type.

conditions of -0.19 . Fig. 1 illustrates the grand mean ERPs for participants in the upper and lower quartiles of MI. There was a significant 3-way interaction between Condition, Site, and NPI. At more anterior sites, NPI was positively related to P3 in the B–X and B–Y conditions, but less so in the A–X or A–Y conditions. At Pz, however, relationships were smaller and occasionally negative. Fig. 2 illustrates the grand mean ERPs for participants in the upper and lower quartiles of NPI. No other effects were statistically significant.

3.3. N2 Amplitude

A parallel mixed model was fit for N2 amplitude. Table 3 presents the *F*-statistics and *p*-values for these analyses. There was a significant main effect for Condition. Amplitude in the A–X condition was significantly smaller (i.e., less negative) than in the A–Y ($p = 0.006$), B–X ($p < 0.001$), or B–Y ($p < 0.001$) conditions. There was also a significant Site main effect for N2 amplitude to be more negative the more

Table 4
Standardized partial regression coefficients derived from fixed effect estimates in mixed models.

		P3 amplitude				N2 Amplitude				RT	Response accuracy
		Fz	Cz	Pz	Mean	Fz	Cz	Pz	Mean		
MI	A–X	–0.17	–0.21	–0.22	–0.17	–0.23 ^b	–0.21 ^b	–0.10 ^b	–0.15	0.00	–0.19
	A–Y	–0.23	–0.20	–0.13	–0.18	–0.23 ^b	–0.20 ^b	–0.12 ^b	–0.17	0.07	–0.25
	B–X	–0.24	–0.26	–0.21	–0.22	–0.31 ^b	–0.38 ^b	–0.25 ^b	–0.26	0.02	0.04
	B–Y	–0.25	–0.28	–0.20	–0.22	–0.21 ^b	–0.23 ^b	–0.21 ^b	–0.18	0.03	–0.01
	Mean	–0.21	–0.23	–0.17	–0.19 ^a	–0.24	–0.25	–0.17	–0.19 ^a	0.03	–0.08
NPI	A–X	0.05 ^b	0.13 ^b	0.05 ^b	0.07	0.11 ^b	0.18 ^b	0.19 ^b	0.13	0.07	0.13
	A–Y	0.17 ^b	0.08 ^b	–0.07 ^b	0.07	0.26 ^b	0.16 ^b	0.10 ^b	0.16	0.08	0.26
	B–X	0.23 ^b	0.21 ^b	0.10 ^b	0.17	0.07 ^b	0.14 ^b	0.10 ^b	0.08	0.26	0.09
	B–Y	0.25 ^b	0.24 ^b	–0.05 ^b	0.14	0.12 ^b	0.13 ^b	0.13 ^b	0.11	0.16	–0.06
	Mean	0.16	0.16	0.00	0.10	0.15	0.15	0.12	0.12	0.12	0.08
AI	A–X	0.05	0.00	0.06	0.03	–0.07 ^b	–0.22 ^b	–0.23 ^b	–0.14	0.02	0.03
	A–Y	–0.07	–0.06	0.05	–0.03	–0.12 ^b	–0.14 ^b	–0.09 ^b	–0.10	–0.01	0.14
	B–X	0.03	0.00	0.05	0.02	0.15 ^b	0.15 ^b	0.20 ^b	0.14	–0.15	–0.10
	B–Y	0.04	0.05	0.11	0.06	0.10 ^b	0.05 ^b	0.05 ^b	0.06	–0.11	0.05
	Mean	0.01	0.00	0.06	0.02	0.02	–0.03	–0.01	–0.01	–0.06	0.02
Sub. Use	A–X	–0.01	–0.01	–0.03	–0.01	0.21	0.24	0.06	0.14	0.00	–0.08
	A–Y	0.06	0.11	0.07	0.07	0.04	0.12	0.09	0.07	0.17	–0.24 ^d
	B–X	0.16	0.11	0.06	0.10	0.23	0.24	0.12	0.17	0.14	–0.18 ^d
	B–Y	0.03	–0.03	–0.06	–0.02	0.18	0.17	0.06	0.12	0.10	–0.29 ^d
	Mean	0.05	0.04	0.01	0.03	0.15	0.19	0.08	0.12	0.09	–0.19 ^c

Note: MI = Barratt Impulsiveness Scale 11 (BIS-11) Motor Impulsiveness; NPI = BIS-11 Non-Planning Impulsiveness; AI = BIS-11 Attentional Impulsiveness; Sub. Use = substance use is the first principal component from a principal component analysis of substance use measures. All continuous predictor variables were sex-corrected and were the unstandardized residuals from regressions using dummy coded sex as a predictor. Sex is dummy coded (0 = male, 1 = female). a = significant BIS-11 facet main effect in mixed model; b = significant BIS-11 facet × Condition × Site interaction in mixed model; c = significant substance use main effect in mixed model; d = significant substance use × Condition interaction in mixed model.

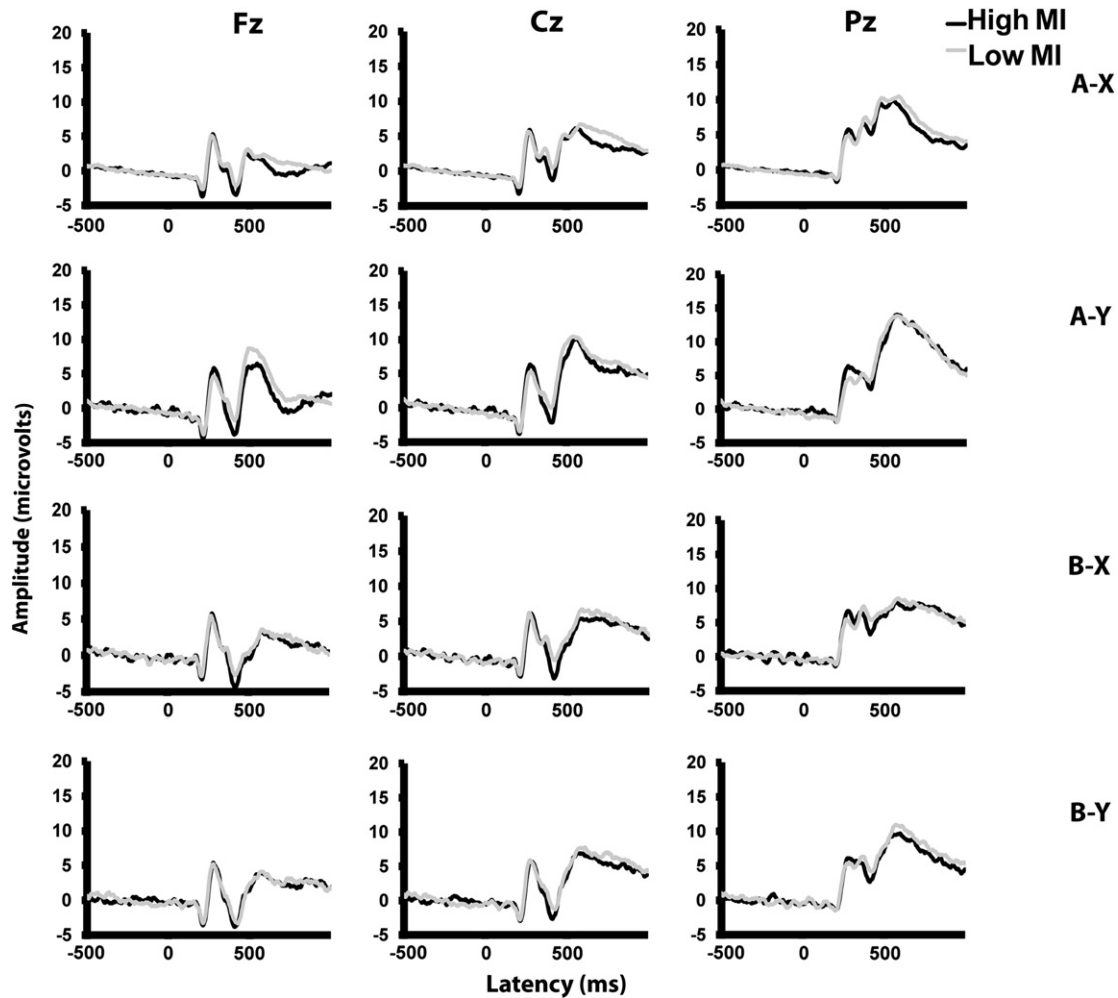


Fig. 1. Grand mean ERP waveforms for participants high (i.e., in the upper quartile) and low (i.e., in the lower quartile) on BIS-11 Motor Impulsiveness (MI) for all task conditions and by electrode site.

anterior the electrode site ($Fz < Cz < Pz$; all p -values < 0.001). A significant Condition $\times$ Site interaction was identified that reflected a trend for a greater difference between conditions at Cz than at the other sites. Further, there were significant Condition $\times$ Sex and Condition $\times$ Site $\times$ Sex interactions. Only at Cz was there a statistically significant Condition $\times$ Sex interaction, $F(3,85) = 6.02$, $p = 0.001$. At this site, both A–X and A–Y N2 tended to be more negative than B–X and B–Y for men compared to women. No other effects were significant.

There was a significant MI main effect for N2. The average relationship across sites and conditions was $\beta = -0.19$. There were, however, significant MI $\times$ Condition and MI $\times$ Condition $\times$ Site interaction effects. As can be seen in Table 4, MI was associated with larger (i.e., more negative) N2 in all conditions at Fz and Cz but for the Pz scalp site there was only a significant correlation between MI and N2 in the B–X condition. There were also significant AI $\times$ Condition and AI $\times$ Condition $\times$ Site interactions, but no main effect for this scale. As illustrated in Table 4, AI tended to be negatively related to N2 in the A–X and A–Y conditions, but positively in the B–X and B–Y conditions. Fig. 3 depicts the grand mean waveforms for participants in the upper and lower quartiles of AI. Finally, there was a significant NPI $\times$ Condition $\times$ Site interaction. Amplitude was most strongly related to NPI at Fz in the A–Y condition ($\beta = 0.26$).

3.4. RT

A similar model was fit with RT with the exception that there were no Site terms. Results are reported in Table 3. RT differed significantly across conditions. RT was longer in the A–Y condition than in the A–X, B–X, and B–Y conditions (all p -values < 0.001). RT in the A–X condition was also significantly longer than in the B–Y condition, $p = 0.005$. No other effects were significant.

3.5. Response accuracy

Because we eliminated subjects with performance less than 80% correct in the key A–X and A–Y conditions, the number of correct responses was negatively skewed. As such we did an arcsine transformation of the proportion of correct responses and fit a similar model to what we did with RT. Results are reported in Tables 3 and 4. Accuracy differed significantly across task conditions. In particular, accuracy was significantly lower in the A–Y condition than in the A–X, B–X, or B–Y conditions (all p -values < 0.001). Accuracy was also significantly greater in the A–X than the B–Y condition ($p = 0.003$) and in the B–Y than the B–X condition ($p < 0.001$). There were no other significant differences among conditions. Women had significantly less accurate performance than men. There was a significant main effect for substance use and a significant Condition $\times$ substance use interaction. Substance use was

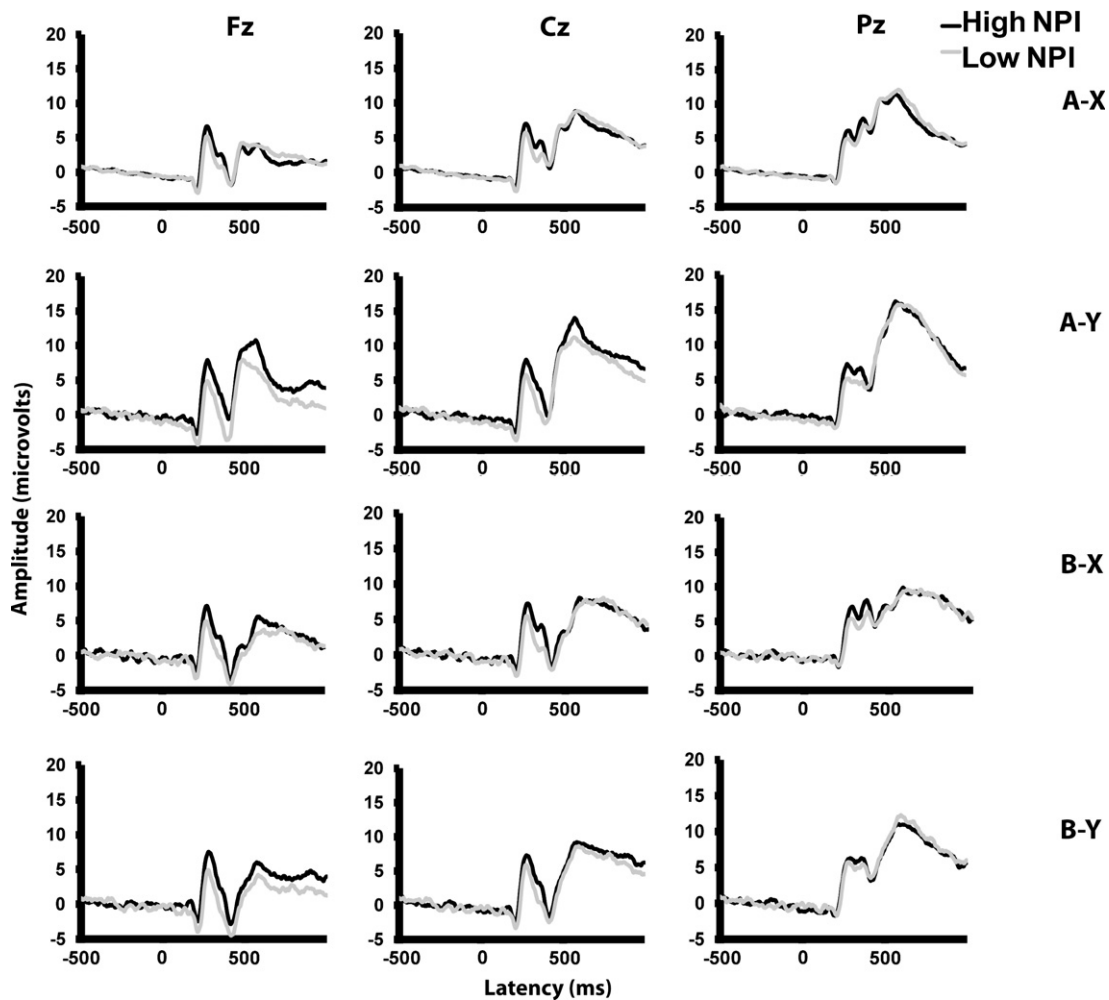


Fig. 2. Grand mean ERP waveforms for high (i.e., in the upper quartile) and low (i.e., in the lower quartile) on BIS-11 Non-Planning Impulsiveness (NPI) for all task conditions and by electrode site.

more associated with poor task performance in the A-Y ($\beta = -0.24$) and B-Y ($\beta = -0.29$) conditions than in the A-X ($\beta = -0.08$) and B-X ($\beta = -0.18$) trials.⁴

⁴ Despite our efforts to reduce the effects of sex on our measures prior to analysis, it could be argued that our results may stem from the imbalance of participants of each sex in our sample. As such, we selected a subset of 25 men and 25 women matched on ethnic background (56% East Asian, 36% European, 8% other) and age ($M = 19.95$ years, $SD = 1.24$). We reran all analyses with the non-sex corrected data from this subset of participants only. Results involving BIS-11 facets and substance use are briefly described in what follows. For P3 amplitude, there were no longer significant effects involving NPI, however, not only was the relationship with MI robust in this sex-balanced subsample the magnitude of the relationship increased, $F(1,50) = 7.78$, $p = 0.007$, $\beta = -0.31$. For N2 amplitude, although there was still a marginal relationship between MI and N2 amplitude, it was no longer statistically significant, $F(1,50) = 3.78$, $p = 0.058$. A similar trend towards attenuated findings in this restricted sample was found for the MI $\times$ Condition interaction, $F(3,50) = 2.26$, $p = 0.092$. There were no longer even marginal effects involving NPI. The AI $\times$ Condition interaction remained significant, $F(3,50) = 4.38$, $p = 0.008$. This reflected a generally larger positive relationship between N2 amplitude and AI in the B-X condition ($\beta = 0.16$) than in the A-Y ($\beta = 0.02$) or B-Y ($\beta = 0.04$) conditions. The relationship was negative in the A-X condition ($\beta = -0.09$). Substance use was still significantly related to response accuracy, $F(1,50) = 5.71$, $p = 0.021$, $\beta = -0.21$. The interaction between substance use and condition was still statistically significant for accuracy, $F(3,50) = 2.96$, $p = 0.041$. Substance use was only significantly related to response accuracy in the B-X condition ($\beta = -0.37$). Our findings with MI, AI, and substance use appear robust, and are present at significant or marginally significant levels in this subsample. Relationships involving NPI appear to be less robust. Interactions between sex and the condition factor remained for P3 and N2.

3.6. BIS-11 total scores

Part of our rationale for conducting this study was the idea that facets of the BIS-11 may have differential relationships with aspects of cognitive task performance and that this heterogeneity may mask findings if only the BIS-11 total score is examined. We conducted each set of analyses with a sex-corrected BIS-11 total score in the place of the subscale scores. Substance use and sex were included in analyses as before. There were no statistically significant main effects, 2-way or 3-way interactions involving BIS-11 total score and P3 (p -values ≥ 0.189) or N2 (p -values ≥ 0.250) amplitude. Similarly, there were no significant effects involving BIS-11 total scores and RT (p -values ≥ 0.339) or Accuracy (p -values ≥ 0.430).

4. Discussion

This is the first study to investigate the specific relationships between the three facets of BIS-11 impulsivity and executive processes. We used an expectancy AX-CPT and found that the A-Y condition was associated with longer RT and poorer accuracy consistent with this condition taxing inhibitory control. We found the expected P3 amplitude patterns across task conditions and scalp sites. A trend for relatively greater P3 amplitude in the A-Y condition at the frontal scalp site compared to other conditions and electrode locations was detected. These findings suggest a component similar to a No-Go P3 (Fallgatter et al., 1997; Kirmizi-Alsan et al., 2006; Smith et al., 2008; Tekok-Kilic et al., 2001) being generated in the A-Y condition as

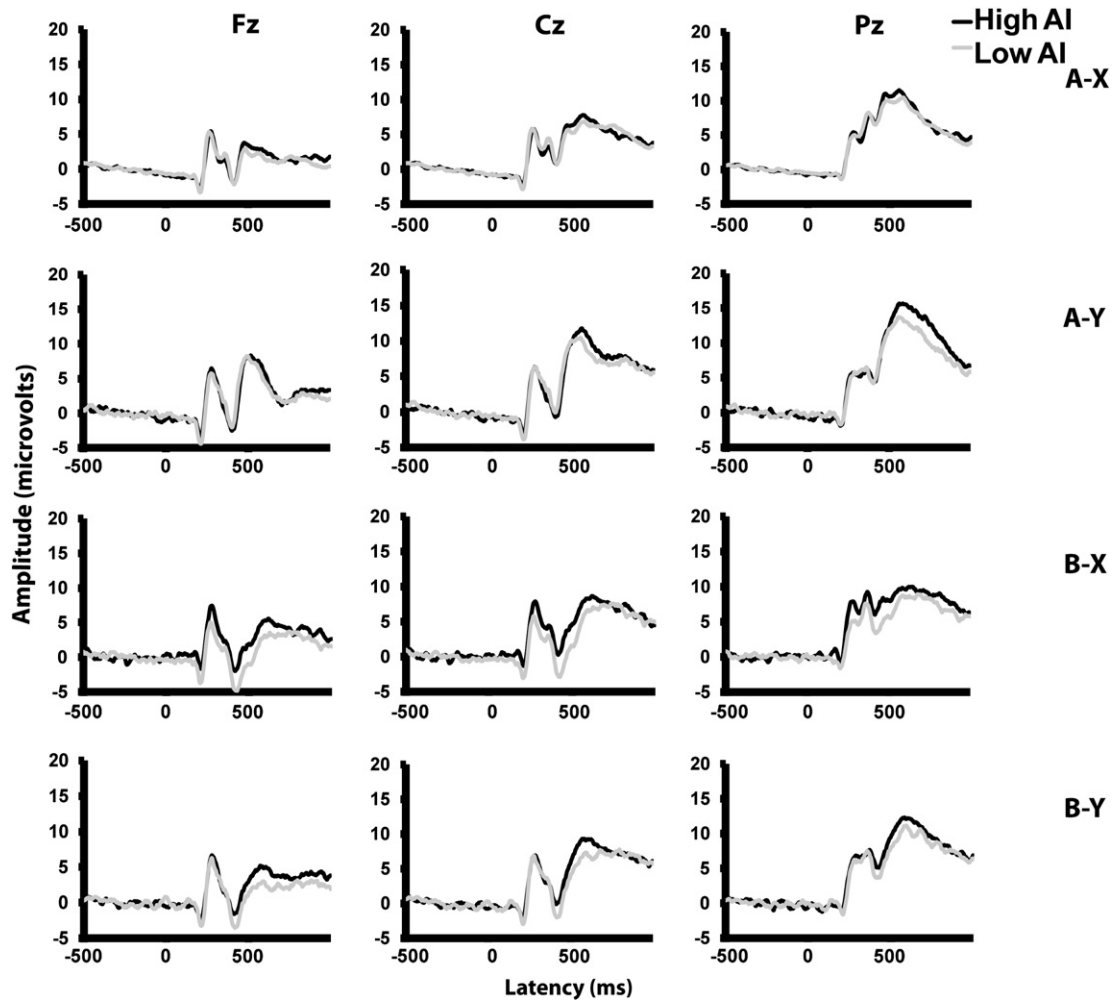


Fig. 3. Grand mean ERP waveforms for high (i.e., in the upper quartile) and low (i.e., in the lower quartile) on BIS-11 Attentional Impulsiveness (AI) for all task conditions and by electrode site.

measured at frontal scalp sites. Greater N2 amplitude at frontocentral sites and in conditions requiring the non-prepotent response consistent with engagement of conflict detection processes necessary for inhibition (Falkenstein, 2006; Nieuwenhuis et al., 2003) was observed. Unlike in a standard Go/No-Go task in which no response is made on the No-Go trials, in the AX-CPT the primed response is not only inhibited, but an appropriate alternative is substituted. As such, the AX-CPT is not a true Go/No-Go task, but the response inhibition and conflict detection demands of some conditions, notably A-Y, likely influences P3 and N2. It is possible that the P3 and N2 seen in such a task reflect a mix of inhibitory and conflict detection process shared with Go/No-Go tasks as well as other influences.

Although latency was not a focus of the present study, it is interesting that P3 latency appeared to be longest in the B-X and B-Y conditions and longer than behavioral RT in all conditions other than A-Y. For those with reasonably intact working memory, the conditions with the B cue do not require full identification of the probe stimulus prior to response selection, allowing for a response to be initiated prior to fully processing the stimulus. This is in contrast to the A-Y condition which requires full processing of the probe stimulus for execution of the correct response.

We conducted mixed model analyses to examine the unique variance in behavioral and ERP measures accounted for by each facet of the BIS-11. Three sets of findings emerged. First, N2 amplitude increased with levels of MI and this was most pronounced in the B-X condition; however P3 amplitude decreased in all conditions as MI levels increased. Second, individuals high on NPI displayed lower N2

in the A-Y condition at frontal sites only, and relatively greater frontocentral P3 amplitude in the B-X and B-Y conditions. Third, higher AI was most strongly associated with increased N2 amplitude in the A-X condition. These findings occurred above and beyond the effects of substance use, and they hold for partial relationships with the facets. BIS-11 total scores were unrelated to any AX-CPT measure, highlighting the importance of examining sub-traits. Taken together, our results support differential relationships between the three facets of the BIS-11 and various cognitive processes. Here, we first discuss the implications of our findings for each subscale. We then consider limitations and potential future directions.

4.1. Motor impulsiveness

MI is characterized by a lack of behavioral control and acting quickly without adequately considering consequences. Given the pattern of chronic rashness leading to potentially reckless behavior reflected by MI and tentative findings of greater relationships between MI and behavioral measures of disinhibition (Spinella, 2004; Gorlyn et al., 2005; Keilp et al., 2005), we hypothesized that MI would be related to reduced N2 and P3 amplitudes in the A-Y condition, reflecting impaired inhibition and conflict detection necessary for response suppression. Reduced P3, however, was associated with unique variance in MI across sites and conditions. As our finding was not specific to any condition it may reflect a general limitation on cognitive processes contributing to P3 across levels of inhibitory demand in individuals high

in MI that could undermine performance on a variety of tasks, not just those involving response suppression.

Contrary to our hypothesis, higher levels of MI were associated with greater, not reduced N2 amplitude and this was most pronounced in the B–X condition. It may be that this condition represented a greater challenge for those high on MI than for those lower on this trait. Arguably, it is a condition creating a heightened challenge for conflict detection. The X-probe is typically responded to with a target response on the much more frequent A–X condition, and the B–X condition is the only condition that requires the non-target response be produced to the stimulus most often associated with that prepotent response. In contrast, the A–Y and B–Y conditions involve a non-target response to a probe stimulus that is never associated with the target response. Although in isolation, the greater N2 associated with MI in the B–X condition could reflect greater inhibition in this condition by individuals high on MI, it should be noted that we did not detect a similar trend for greater inhibition in this condition as indicated by greater anterior P3 amplitude being related to higher MI. In fact, reduced, not augmented, P3 was associated with MI. It may be that compared to other conditions, the B–X condition evokes greater response conflict in those high on MI because of the mismatch between the prepotent response to the probe (X) and the required response established by the cue (B). Thus, these individuals may engage greater conflict detection resources in order to maintain an adequate level of task performance. It should also be noted that MI had the most restricted range of the three scale scores, as well as having modest internal consistency, and therefore P3 and N2 relationships with this scale are likely underestimates.

4.2. Non-planning impulsiveness

NPI reflects distaste for engaging in tasks involving mental complexity and cognitive engagement in planning. Contrary to our hypothesis, despite a trend for an $NPI \times \text{Condition}$ interaction, there was no significant relationship between RT and NPI. High levels of NPI were, however, associated with greater frontal P3 amplitude in the B–X and B–Y conditions. Increased frontal P3 amplitude suggests that individuals high on NPI may require relatively more inhibition of motor responses to probes following the B-cues. For the A–X condition the prepotent target response requires no inhibition and poses little challenge for individuals at all points on the NPI dimension. On the other hand, the A–Y condition requiring inhibition of a primed response may tax the inhibitory resources of those across the bulk of the NPI continuum. The B–X and B–Y conditions may be intermediate in challenging inhibitory control and may not be as subject to floor and ceiling effects. In order to maintain adequate performance on these conditions those high on NPI may need to inhibit the more commonly occurring response more than those lower on the trait. Perhaps, low NPI individuals use context provided by the cue to more efficiently respond on these conditions.

Higher NPI was associated with reduced frontal N2 amplitude, but only in the A–Y condition. The A-cue in this condition primes participants to make the common target response to an X, a prepotent response that needs to be inhibited when the probe is a Y. As such, our finding, in the absence of a relationship between anterior P3-indexed inhibition and NPI in the A–Y condition, suggests that individuals high on this trait may do a poorer job detecting conflict when a prepotent response is primed by a cue. Complex mental activities involving detection of conflicts between response options may be relatively more difficult for individuals high on this trait. In turn, this may lead to an avoidance of planning and complex mental activity in daily life, particularly when planning requires evaluating and choosing between multiple and conflicting behavioral options.

4.3. Attentional impulsiveness

AI is characterized by restlessness, a tendency towards distraction, and an inability to maintain sustained, focused attention. We found that AI did not significantly account for any unique variance in P3 amplitude. However, individuals higher on AI had relatively enhanced N2 in the A–X condition. As N2 is expected to be greater in conditions that require the inhibition of a prepotent response, but the A–X condition requires the implementation of a prepotent response, this may reflect a relatively inefficient conflict detection system in those high on AI. That is, the executive resources of conflict detection have been allocated to a task condition that does not require it. There was little evidence of a relationship between AI and N2 in the other conditions, so the significant relationship detected in the A–X condition may reflect a tendency to engage conflict detection relatively consistently in that task condition specifically.

4.4. Substance use

Substance use is a widely recognized correlate of BIS-11 impulsivity (e.g., [Stanford et al., 2009](#)) and MI, in particular ([Carlson et al., 2010](#)). In our study, substance use was associated with significantly worse task performance, but not with significant ERP effects. It is important to note that the ERP findings with BIS-11 scales involved individual differences in impulsive traits not shared with our substance use measure. This suggests that relationships between these impulsive traits and executive processes cannot be accounted for by substance use alone.

It is worth noting that substance use was not as strongly correlated with BIS-11 scales in our sample as in other samples (e.g., [Stanford et al., 2009](#)). Although our composite substance use measure was fairly normally distributed about 1/3 of our sample had not drank more than 7 drinks in the past year, perhaps making this sample unusually low in terms of college drinking (cf. [Carlson et al., 2010](#)). The majority of our sample was of East Asian descent and people of East Asian ancestry often report lower alcohol consumption rates than people with European family backgrounds (e.g., [Tu and Israel, 1995](#); [Tucker et al., 2003](#)). This was true of our subjects, as well. An allele associated with lower levels of aldehyde dehydrogenase, which leads to an unpleasant flushing response when alcohol is consumed, is more common in East Asian than European populations ([Tu and Israel, 1995](#)). This along with cultural socialization effects ([Tu and Israel, 1995](#)) potentially contributed to the lower drinking in this sample. Nonetheless, relationships between substance use and AX-CPT measures are likely conservative underestimates because of the truncated range of alcohol use. The lower levels of drinking reduce the chances that our findings of relationships between BIS-11 subscales and measures of inhibition and conflict detection are because of the confounding effects of alcohol. Further, when we did find a relationship between substance use and response accuracy it is likely a robust finding.

Although we did not detect a significant partial relationship between N2 amplitude and substance use, there was a non-significant trend for a positive association between the N2 and the variance in substance use not related to BIS-11 facets. If this proved significant in a sample with greater power, it would be in contrast to the negative relationship between MI and N2 amplitude. Other biopsychological systems, largely conceptually independent from MI have been proposed as influencing substance use such as those involving emotional regulation ([Whiteside and Lynam, 2001](#); [Cyders and Smith, 2008](#)), reward sensitivity or punishment insensitivity (e.g., [Gray and McNaughton, 2000](#); [Dawe and Loxton, 2004](#)), and cortical under-stimulation leading to sensation seeking (e.g., [Zuckerman, 2007](#)). It may be that individual differences in one or more of these variables is a more prominent influence on variance in substance use in this sample and it may be variance in these phenomena,

rather than BIS-11 impulsivity, that is reflected in the partial relationships between N2 amplitude and substance use.

4.5. Behavioral measures

Past studies have reported zero-order correlations between BIS-11 facets and measures of executive function including those requiring inhibition (e.g., Spinella, 2004; Gorlyn et al., 2005; Keilp et al., 2005). Yet, we did not find any partial associations between BIS-11 subscale scores and behavioral measures. Our subjects, however, were restricted to those with 80% or better accuracy so restriction of range may have attenuated the magnitude of relationships we could observe between BIS-11 facets and response accuracy. Since the subjects included in our analyses had relatively good performance despite differences in electrophysiological measures, our findings underscore the importance of using measures of brain activity in addition to behavioral measures in elucidating the underpinnings of impulsive traits. It is possible that well-motivated subjects may engage extra resources to compensate for a cognitive limitation in order to maintain an adequate level of performance (see Goya-Maldonado et al., 2010). The relatively larger N2 amplitude associated with MI may reflect such compensation. Presumably this would be energy intensive and difficult to maintain in all contexts and may result in breakdowns in control of behavior outside of the laboratory.

4.6. Limitations and future directions

Our findings should be interpreted with the following considerations. First, the majority of the sample was female. Past research has found that females tended to exhibit lower levels of impulsivity (Chapple and Johnson, 2007; Pompili et al., 2007), although perhaps less so for BIS-11 facets (Stanford et al., 2009). The only difference in impulsivity measures between men and women in our sample, however, was for MI, and this was in the opposite direction with women having slightly higher levels than men. Our data suggest several other gender differences in the AX-CPT measures. There were no significant sex differences, however, in correlations between AX-CPT measures and either substance use or BIS-11 facets in our sample and we conservatively employed a sex correction approach to remove variance due to sex from our dimensional analyses. In addition, analyses of a sex-balanced subset of participants matched for ethnic background and age also supported most of our primary findings involving MI and AI despite a greatly reduced sample size. Further research should be done, however, with larger, more equally balanced samples of men and women in order to adequately evaluate potential higher order interactions between BIS-11 facets, task variables and sex.

Second, that over half of our participants were of East Asian descent raises the issue of generalizability of our findings. In fact, a relatively high Asian student sample composition is not uncommon for many west coast urban areas in North America and is relatively representative of our student body as reported in other studies of personality from this University (cf. Jones and Paulhus, 2011; Williams et al., 2010). Although students of East Asian background had lower levels of substance use than other subjects, there were no significant differences between our two largest ethnic groups (East Asian and European-family background) in the pattern of correlations between AX-CPT measures (i.e., P3 amplitude, N2 amplitude, RT, and response accuracy) and any BIS-11 facet or substance use. This suggests that our findings are likely not seriously impacted by the relatively high proportion of East Asian subjects in our sample. It is nonetheless important for additional research to replicate and extend these results in other samples with more balanced ethnic compositions and larger samples that would enable evaluation of interaction effects that are underpowered in the present sample.

Third, our sample consisted of undergraduates. This meant that the age range of our participants is rather limited and we did not target individuals with clinically significant levels of impulsive behavior problems. As such, our conclusions may only be generalized to other undergraduate students. Young adulthood, however, is a developmental period involving the final maturational stages of many neural systems implicated in executive function (Sowell et al., 2001; Best and Miller, 2010) and undergraduates are particularly prone to some disinhibited behavior problems associated with high BIS-11 facet levels, such as binge drinking (Carlson et al., 2010). Since impulsivity may be viewed along a continuum, it is important to explore underlying mechanisms across the range of this dimension, not solely at the more extreme levels typically found in clinical and forensic samples. Therefore, this is a subgroup of substantive interest in this area as well as one of convenience to university-based researchers.

Fourth, relationships between the ERP components discussed here and anxiety have been previously observed. As such, larger N2 amplitudes have been previously associated with high state and trait anxiety in conditions requiring response inhibition (Righi et al., 2009; Sehlmeier et al., 2010). In regards to No-Go P3, larger amplitudes have been associated with anxiety sensitivity (Sehlmeier et al., 2010) while smaller amplitudes have been associated with high state anxiety and cognitive failures (Righi et al., 2009). As the focus of the present study was on the relationship between trait impulsivity and executive processes, anxiety was not assessed. Although anxiety may contribute to unexplained variance in our study, it is unlikely that anxiety accounts for the observed effects. The conceptualization of anxiety and impulsivity as orthogonal traits guided development of the BIS; items were selected to be uncorrelated with anxiety (as reviewed in Stanford et al., 2009). In support of this conceptualization, a lack of a significant association between trait anxiety and the BIS-10 total or subscale scores has been previously reported (Barratt et al., 1987). Furthermore, equivalent Beck Anxiety Inventory scores have been documented between subjects split into groups based on low and high BIS-11 scores (Mobini et al., 2006).

Lastly, we also only used one task and converging measures of inhibition, attention, and conflict monitoring are needed to clarify our observed relationships. We found reduced P3 amplitude associated with MI. Intriguingly, despite the past association of reduced P3b amplitude and impulsivity on tasks not explicitly tapping inhibition (De Pascalis et al., 2004; Justus et al., 2001; Venables et al., 2011), including some involving the BIS-11 (Chen et al., 2007; Harmon-Jones et al., 1997; Moeller et al., 2004; Russo et al., 2008), recent work has reported greater P3 amplitude among impulsive individuals defined with Eysenck's impulsivity measures during response inhibition (Dimoska and Johnstone, 2007; Lansbergen et al., 2007). Those studies used a Stop Signal task. It and the AX-CPT may differ in the nature of the inhibition involved. On the Stop Signal task, arguably early stages of the response are already underway when the stop signal occurs, whereas, on the AX-CPT information concerning the need to inhibit a response is present at the onset of a probe stimulus. Future work should address how differences in task demands or how impulsivity is conceptualized results in different relationships between impulsivity and ERP measures of inhibition.

4.7. Conclusion

The present study suggests that the three BIS-11 sub-traits are differentially associated with electrophysiological measures of attention, inhibition and conflict monitoring essential for executive functions. No relationships were found with BIS-11 total scores. Our results were somewhat mixed and as our interpretations are somewhat speculative, future research should be done to replicate and expand on our findings. We suggest that MI may be related to reduction of a cognitive process not specific to inhibition and a heightened tendency to detect conflict when required to correctly make a non-

prepotent response to stimuli that normally pull for such a response. NPI may be associated with worse conflict monitoring when inhibiting a primed response and with more inhibition being required for correct behavioral responses when contextual cues would normally rule out a commonly made response. AI may be associated with an over engagement of conflict detection, when the response choice should be fairly stimulus-driven. Our findings highlight the importance of taking a multi-faceted approach to examining impulsivity in studies of cognitive functions and personality.

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